

TRACKER_AUTH_TESTING

Tracker Credentials Live Testing

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Status summary: Live credential-validation tests + challenge for the 5 supported trackers, wired into the integration suite and the challenge bank.

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Overview

Two artefacts live-validate that the merge service can authenticate against each supported tracker using the credentials configured in `.env`:

- `tests/integration/test_tracker_auth_live.py` — a parametrized pytest live integration test, one parameter per tracker.
- `challenges/scripts/tracker_auth_live_challenge.sh` — a bash challenge that drives the same end-to-end auth path.

Both exercise the **real running merge service** (port 7187) and assert on its reported authentication state. They never inspect, print, or log any credential value (constitution §11.4.10).

The five trackers

Tracker	Type	Auth model
rutracker	private	Username/password login (cookies; CAPTCHA may be required periodically)
kinozal	private	Username/password (KINOZAL_*, falling back to IPTORRENTS_*)
nnmclub	private	Cookie-based auth (NNMCLUB_COOKIES)
iptorrents	private	Username/password (IPTORRENTS_*); freeleech-only downloads
rutor	public	No authentication — searches and downloads anonymously

For the four private trackers the test asserts that valid credentials produce an authenticated session. For rutor there is no login endpoint; the test asserts only that it is reachable and needs no auth (a positive auth assertion is intentionally not made).

How the live tests validate credentials

The live test/challenge validates each credential by driving a REAL search through the merge service and reading the per-tracker authentication state the service itself reports during that search:

1. Confirm the merge service is up (GET /health).
2. Start a real search — POST /api/v1/search with a public query ({ "query": "ubuntu", "limit": 50 }) — then poll GET /api/v1/search/{search_id} until the search reaches a terminal state (completed / no_results).
3. The response carries a tracker_stats list — one entry per tracker — each shaped {name, status, authenticated, results_count, error}. For each **private** tracker (rutracker, kinozal, nnmclub, iptorrents) assert authenticated is true AND status is success or empty — proof the stored credentials logged in against the real tracker. For **rutor** (public, no login) assert status == "success" and results_count > 0.

The assertion oracle is the **authenticated flag the merge service reports for the tracker after attempting a real login during the search** — a user-observable outcome of the real auth path — never the mere absence of an error and never any credential string (anti-bluff, constitution §11.4 / §11.4.10). No /api/v1/auth/* status endpoint is consulted; the search response's tracker_stats is the single source of truth.

Honest-SKIP vs FAIL rules

Per constitution §11.4.3 (topology-appropriate dispatch) and §11.4.69 (no fail-open SKIP), the verdicts are:

- **SKIP** — the merge service is **down / unreachable**: the topology required to validate is absent, so the test SKIPS with a reason rather than failing.
- **SKIP** — a **CAPTCHA or transient** blocks login (e.g. RuTracker cookies expired and a CAPTCHA is now required): this is an operator-attended/transient condition, not a product defect, so it SKIPS with a reason.
- **FAIL** — the service is up, no captcha/transient is in play, and the tracker reports `authenticated=false`: the configured credentials are genuinely **bad** (or the auth path is broken). This is a real failure.

A SKIP is never silently counted as a PASS, and a missing/empty authenticated flag is treated as a failure to prove auth, not as proof.

Security: never read or log credential values

- The tests assert **only** on the service-reported `tracker_stats[].authenticated` boolean (plus `status / results_count`). They never read `RUTRACKER_PASSWORD`, `KINOZAL_PASSWORD`, `NNMCLUB_COOKIES`, `IPTORRENTS_PASSWORD`, or any other secret value, and never echo container env values.
- `.env` is never committed (constitution §11.4.10; project `CLAUDE.md`).
- No credential value appears in test output, challenge output, or logs.

How to run

Prerequisite: the merge service must be running on port 7187 with valid credentials present in `.env` (otherwise the tests SKIP, see above).

```
# Pytest live integration test (auto-collected by ci.sh PHASE 4)
python3 -m pytest tests/integration/test_tracker_auth_live.py -v
      --import-mode=importlib
```

```
# The bash challenge (auto-discovered by the challenge aggregator)
bash challenges/scripts/tracker_auth_live_challenge.sh
```

```
# Or as part of the wider suites:
```

```
./ci.sh                                # PHASE 4 collects tests/  
integration/                           # tests/integration +  
scripts/run-tests.sh live              tests/e2e  
bash challenges/scripts/run_all_challenges.sh
```

Where these are wired

- **Pytest collection** — tests/integration/
test_tracker_auth_live.py is auto-collected by ci.sh PHASE 4
(pytest ../tests/integration/) and by scripts/run-tests.sh
live. No collection change was needed (default python_files =
test_*.py; no override in pyproject.toml).
- **Challenge aggregator** — challenges/scripts/
run_all_challenges.sh globs *_challenge.sh in its own directory,
so tracker_auth_live_challenge.sh is auto-discovered there. No
explicit registration was needed.
- **HelixQA bank** — BOBA-SRV-015 in submodules/helixqa/banks/
boba-services.yaml (symlinked into challenges/helixqa-banks/
boba-services.yaml) references both artefacts by path with an
anti-bluff note that the authenticated flag is the assertion oracle
and that no credential value is ever read.